DRAGONSCAN v5.2.0: SYSTEM SPECIFICATION MANUAL

Comprehensive Architectural, Mathematical, Algorithmic & Hardware Reference Guide Multi-Modal 60 GHz FMCW Radar & Computer Vision Station for Dragon Fruit Quality Inspection

Target Host System:	Raspberry Pi 5 (8GB RAM) | Debian 12 Bookworm (64-bit)	**Date of Release:**	October 2026
Radar Sensor:	Hi-Link HLK-LD6002 (VERC V1.0) 60 GHz Millimeter-Wave FMCW	**Carrier Board:**	ANDAR ADTE210400XXNA01L V2.0 (20230828) USB-C
Optical Sensor:	Logitech C270 HD USB Webcam (720p @ 30 FPS MJPG)	**Actuator Interface:**	RP1 GPIO 17 Sorter Diverter / Relay Pulse

1. Executive Summary & Problem Formulation

Post-harvest sorting of red and white pitahaya (dragon fruit, *Hylocereus undatus / polyrhizus*) faces a severe latent defect challenge: **subsurface translucent pulp decay and internal softening**. While external anthracnose and mechanical bruising manifest on the outer rind, early-to-moderate fungal breakdown and water maceration frequently leave the exterior bracts and peel visibly sound. Traditional optical camera grading (computer vision alone) suffers false acceptances of internally ruined fruit. Conversely, manual destructive penetrometry destroys market value.

DragonScan v5.2.0 resolves this defect by establishing a **synchronized multi-modal sensing architecture**: (1) A 60 GHz Millimeter-Wave FMCW radar module that non-destructively penetrates the rind to probe internal pulp water content and dielectric permittivity (ε_r), and (2) an optical camera bench that computes peel necrosis, bract greenness, and morphometric surface area. Data streams are fused via an autonomous Bayesian heuristic engine that commands physical pneumatic segregation.

2. Complete Hardware Bill of Materials (BOM) & Physical Rig

The system operates on industrial agritech hardware calibrated for an inspection chamber with a 250 mm focal height:

Subsystem	Component & Model	Hardware Specifications & Physical Role
Compute Host	Raspberry Pi 5 (8GB RAM)	Broadcom BCM2712 Quad-Core ARM Cortex-A76 @ 2.4 GHz. Executes Python 3 multi-threaded GUI, OpenCV color segmentation, and serial processing.
Radar Transceiver	Hi-Link HLK-LD6002 (VERC V1.0 Daughterboard)	60.0 GHz center frequency (58–62 GHz sweep, 4.0 GHz bandwidth). Contains dual microstrip patch antenna arrays (Tx and Rx). Penetrates fruit tissue 15–30 mm.
Radar Carrier Board	ANDAR ADTE210400XXNA01L V2.0 (20230828)	USB-C interface board housing U1 USB-to-UART bridge IC and U2 ultra-low-noise 3.3V LDO regulator. Connects radar TXD/RXD to Raspberry Pi USB port at 115200 baud.
Optical Camera	Logitech C270 HD Webcam (USB 2.0 Optical Bench)	1280x720 (720p) resolution @ 30 FPS MJPG streaming. Rigidly positioned vertically 250 mm directly above fruit cradle with locked exposure & white balance.
Touchscreen Cockpit	7-inch DSI / HDMI LCD (1024x600 Capacitive Touch)	High-contrast blueprint dark dashboard. Displays live Range-FFT oscilloscope, circular progression dial, and real-time accuracy benchmarks.

Pneumatic Sorter	RP1 GPIO 17 Relay / Actuator Diverter Solenoid	12V 2A pneumatic solenoid diverter arm. Triggered by active-high pulse (0.35s duration) when fruit exceeds segregation rot thresholds.
Power & Thermal	Raspberry Pi 27W USB-C PD + Official Active Cooler	Delivers stable 5.1V / 5.0A power delivery across all USB bus peripherals while maintaining CPU core thermals below 52°C under sustained 30 FPS workloads.

3. Python Imports, Modules & Architectural Justifications

To ensure zero-dependency field resilience across Debian Bookworm (PEP 668 externally managed environment) and Windows, the codebase eliminates unnecessary dependencies while maintaining strict modular separation:

Module / Import	Source Library	Engineering Role & Justification in DragonScan
cv2	OpenCV (headless/contrib)	Core image acquisition, frame resizing, BGR to HSV/LAB color transformations, morphological noise filtering, and in-memory PPM byte stream encoding.
numpy (np)	NumPy Core	Matrix operations, 1D Range-FFT computation (np.fft.rfft), Hanning windowing, peak power dB conversion, and statistical color masking.
serial, glob	PySerial & Python standard	Universal multi-port auto-discovery (/dev/ttyUSB*, /dev/ttyACM*) and packet frame parsing for the ANDAR HLK-LD6002 radar.
tkinter, ttk	Tkinter (Standard Python)	Native GUI desktop runtime. Commands zero overhead on Raspberry Pi X11/Wayland sessions without requiring Chromium/Electron bloat.
threading, queue	Python Concurrency	Separates 30 FPS video streaming and 1.2s radar chirp background polling from the main UI thread, preventing GUI stutter or latency spikes.
pathlib.Path	Python Standard	Cross-platform filesystem normalization across Windows backslashes and Linux forward slashes for dataset directory resolution.
RPI.GPIO / lgpio	Raspberry Pi GPIO	Direct hardware pin manipulation for GPIO 17 ejection relay on Raspberry Pi 5 RP1 southbridge with synthetic mock fallback on PC.

4. Mathematical Formulations & Physical Logic Foundations

A. FMCW Radar Intermediate Beat Frequency (f_b)

The Hi-Link HLK-LD6002 radar emits linear frequency-modulated chirps starting at carrier frequency $f_c = 60.0\ GHz$ sweeping across bandwidth $B = 4.0\ GHz$ (4000 MHz) over sweep duration $T_{sweep} = 400\ \mu s$. The chirp sweep slope S is defined as:

$$\mathbf{S = B / T_{sweep} = 4.0 \times 10^{9}\ Hz / (400 \times 10^{-6}\ s) = 1.0 \times 10^{13}\ Hz/s}$$

When microwave radiation reflects off the dragon fruit placed at target distance R (meters), the round-trip propagation time is $\tau = 2R / c$ (where $c = 3.0 \times 10^{8}\ m/s$). Mixing the transmitted chirp with the received echo yields the intermediate beat frequency f_b:

$$\mathbf{f_b = 2 \cdot S \cdot R / c = 2 \cdot (B / T_{sweep}) \cdot R / c}$$

For a standard chamber inspection height of $R = 25\ cm = 0.25\ m$:

$$\mathbf{f_b = (2 \cdot 10^{13} \cdot 0.25) / (3.0 \times 10^{8}) = 5.0 \times 10^{12} / (3.0 \times 10^{8}) \approx 16.67\ kHz}$$

This 16.7 kHz beat frequency falls precisely inside the audio/IF sampling band of the ADTE2104 baseband ADC, allowing precise Range-FFT distance and energy resolution.

B. Subsurface Dielectric Constant (ε_r) & Internal Flesh Degradation

Healthy firm pitahaya flesh consists of organized cellular pectin and bound moisture, presenting an initial dielectric constant $\varepsilon_r \approx 62.0$. As maceration and pectinase enzymes break down cellular walls during rot, free unbound water molecules increase while microwave backscatter coherence decreases. The estimated dielectric permittivity is governed by:

$$\varepsilon_r = 62.0 + (I_{decay} \times 16.5) \quad (\varepsilon_r > 74.0 \text{ indicates severe waterlogged rot})$$

The internal decay index $I_{decay} \in [0.0, 1.0]$ is derived from the Range-FFT spectral peak sharpness and energy dissipation ratio:

$$I_{decay} = 1.0 - \text{clip}((E_{peak} / \sum E - 0.12) / 0.15, 0.0, 1.0)$$

C. Computer Vision Peel Necrosis Quantification (HSV Segmentation)

Dragon fruit peel features three distinct color channels in HSV space:

- **Ripe Magenta Peel:** Hue $[140, 180] \cup [0, 10]$, Saturation $[45, 255]$, Value $[60, 255]$
- **Bract Foliage Green:** Hue $[32, 85]$, Saturation $[35, 255]$, Value $[40, 255]$
- **Rot/Necrotic Black-Brown Lesions:** Hue $[0, 30] \cup [150, 180]$, Saturation $[30, 255]$, Value $[15, 80]$

The rotten peel percentage is calculated with pixel-perfect precision over the total fruit contour area:

$$\text{Rotten Area \%} = (N_{rot_pixels} / N_{fruit_pixels}) \times 100.0$$

D. Multi-Modal Decision Fusion Boundary

The segregation decision score S_{final} fuses optical peel necrosis with radar internal decay using calibrated weights ($w_v = 0.65$, $w_r = 0.35$):

$$S_{final} = (w_v \cdot \text{Rot\%} / 15.0) + (w_r \cdot I_{decay} / 0.35)$$

If $\text{Rot\%} \geq 15.0\%$ OR $I_{decay} \geq 0.35$ OR $S_{final} \geq 1.0$, the actuator relay on GPIO 17 fires immediately to eject the fruit.

5. Complete Functions, Methods & Class Blueprint

The following matrix documents the exact primary functions, signatures, inputs, and return types across all modules:

Module & Class	Function / Signature	Operational Logic & Return Values
RadarController (hardware/radar_controller.py)	initialize() -> bool	Scans all candidate ports (/dev/ttyUSB*, /dev/ttyACM*) across 5 baud rates. Sends wake probe sequence 0xFD 0xFC 0xFB 0xFA. Binds physical sensor.
RadarController (hardware/radar_controller.py)	scan(rot, angle) -> Dict	Executes FMCW chirp sweep. Reads physical UART packets from ANDAR board. Returns carrier frequency (60.0 GHz), beat frequency (16.7 kHz), distance, decay index, and spectrum.
RadarController (hardware/radar_controller.py)	_read_physical_radar(...)	Parses HLK binary frames (0xFD 0xFC 0xFB 0xFA), ASCII reports, and raw energy bytes into target distance (mm) and power (dB). Aligns Range-FFT spectrum.
CameraController (hardware/camera_controller.py)	get_frame() -> Tuple[bool, ndarray]	Acquires real-time 720p 30 FPS frame from Logitech C270 USB device (/dev/video0) via OpenCV VideoCapture with MJPG fourcc decode.
GpioSorter (hardware/gpio_sorter.py)	trigger_desegregation(reason)	Activates Raspberry Pi 5 RP1 GPIO 17 high for 350 ms to energize pneumatic solenoid valve and physically divert degraded fruit into cull bin.
VisionAnalyzer (core/vision_analyzer.py)	analyze_frame(bgr, thresh) -> Dict	Full pipeline: extracts fruit mask, dimensions (mm), surface area (cm²), color distribution (ripe magenta vs green), and calculates rotten peel percentage.
FusionClassifier (core/fusion_classifier.py)	evaluate(vision, radar) -> Dict	Fuses optical and microwave features into composite defect score. Assigns market quality grade (Grade A, Grade B, Reject) and sets is_desegregate flag.
AccuracyTester (core/accuracy_tester.py)	calculate_live_accuracy(...) -> Dict	Computes dynamic confusion matrix (True Positives, False Positives, True Negatives, False Negatives), sensitivity, and overall accuracy on live dataset captures.

DatasetManager (core/dataset_manager.py)	save_capture(...) -> Dict	Stores raw image, annotated overlay, and unified JSON metadata file containing all optical, radar, and fusion metrics into RIPE, ROTTEN, or UNRIPE folders.
DragonScanApp (gui/dashboard.py)	_video_stream_worker()	30 FPS camera loop with concurrent 1.2s background radar sensing. Pushes FRAME and RADAR_UPDATE messages to thread-safe UI queue.
DragonScanApp (gui/dashboard.py)	_run_scan_pipeline_worker()	Coordinates formal 2-stage synchronized scan: (1) Radar sweep with radial dial animation, (2) Camera exposure lock, fusion verdict, and SweetAlert dialog.

6. Step-by-Step End-to-End Operational Workflow

The inspection lifecycle executes through 6 deterministic, automated phases:

Step 1: System Boot & Multi-Port Sensor Enumeration

When ./run_dragonscan.sh or main.py initiates, the station loads configuration tokens from config.py. The camera controller probes `/dev/video0` through `/dev/video4` to lock onto the Logitech C270 USB stream at 720p 30 FPS. Concurrently, the radar controller auto-scans `/dev/ttyUSB*` and `/dev/ttyACM*`, sends the HLK-LD6002 wake packet, and verifies bi-directional serial communication at 115200 baud.

Step 2: Real-Time Standby & Dual-Modal Background Sensing

The UI displays an unobstructed live optical preview with corner reticles and focal target lock. Unlike basic vision systems, the background worker polls the 60 GHz radar every ~1.2 seconds. The Range-FFT spectrum oscilloscope continuously pulses and displays live carrier (60.0 GHz) and intermediate beat frequency (16.7 kHz) telemetry.

Step 3: Fruit Presence Detection & Stabilization

When an operator places a dragon fruit into the 45° angled inspection cradle, the optical segmentation engine detects the fruit contour (exceeding 1500 pixels). In Manual Mode, the operator clicks 'CLICK TO SCAN FRUIT'. In Auto-Scan Mode, the station waits for 800 ms of spatial stability before automatically launching the sequential pipeline.

Step 4: Synchronized Sequential 2-Stage Scan

• **Stage 1 (0% $\rightarrow$ 50% Dial): 60 GHz Microwave FMCW Radar Penetration.** The radar transceiver emits 64 chirps. Reflection signals pass through the peel to probe inner flesh elasticity and water dissipation, producing the internal decay index and dielectric permittivity (ε_r).
• **Stage 2 (50% $\rightarrow$ 100% Dial): Optical Exposure Lock & Morphology.** The camera freezes an uncompressed BGR snapshot. The vision analyzer measures length, width, surface area, and computes the exact percentage of peel necrosis using HSV color space thresholding.

Step 5: Multi-Modal Fusion, Quality Grading & SweetAlert Modal

The fusion engine combines optical peel rot % and radar subsurface decay into a unified defect score. A SweetAlert modal appears on screen detailing the complete verdict: RIPE (Sound), UNRIPE (Immature), or ROTTEN (Defect Ejected).

Step 6: Hardware Ejection & Dynamic Accuracy Logging

If the fruit is classified as ROTTEN, the controller pulses GPIO 17 to energize the pneumatic solenoid relay, physically rejecting the degraded fruit. The capture is auto-saved into the dataset, dynamically updating the confusion matrix and live model accuracy.

7. Step-by-Step Deployment & Execution Guide

A. Raspberry Pi 5 Deployment (Debian Bookworm)

1. Open LXTerminal on Raspberry Pi OS Desktop.
2. Navigate to project directory and pull latest verified firmware:
   cd /home/pi/Documents/dragonfruit && git pull origin main
3. Grant execution permissions to station launcher:
   chmod +x run_dragonscan.sh
4. Plug in the **ANDAR USB-C radar cable** and **Logitech C270 USB webcam**.
5. Launch DragonScan Station Controller:
   ./run_dragonscan.sh

Note: The launcher automatically validates serial permissions (`chmod a+rw /dev/ttyUSB*`) and configures a self-healing virtual environment if system packages are isolated under Debian PEP 668.

B. Windows PC Testing & Simulation Deployment

1. Double-click run_dragonscan.bat in the repository root, or open PowerShell:
   cd C:\Users\paul\Documents\dragonfruit
   python main.py
2. The window will open in 100% full-screen station mode. Press **F11** to toggle fullscreen, or double-click anywhere on the camera canvas.

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